AI 3000/7010: Reinforcement Learning
Tutorial 5

1 Higher Order Markov Policies

Consider a first-order MDP $(\mathcal{S}, \mathcal{A}, P, \mathcal{R})$.

(a) **Second-order policy.**

Suppose that the policy depends on the last two states:

$$\mathbb{P}^\pi(A_n = a \mid S_n = s, A_n = a, S_{n-1} = \tilde{s}, A_{n-1} = \tilde{a}, \dots, S_0 = s_0, A_0 = a_0) = \pi(a \mid s, \tilde{s}).$$

Prove that $\{S_n\}_{n\geq 0}$ process is a second-order Markov chain. That is, show that

$$\mathbb{P}(S_{n+1} = s' \mid S_n = s, S_{n-1} = \tilde{s}, \dots S_0 = s_0) = P^\pi(s' \mid s, \tilde{s})$$

for an appropriate transition kernel P^π, and explicitly derive this kernel.

(b) **General k-th order policy.**

Suppose now that the policy depends on the last k states:

$$\mathbb{P}(A_n = a \mid S_{0:n}, A_{0:n-1}) = \pi(a \mid S_{n-k+1:n}).$$

Show that the state process is an k-th order Markov chain. Derive an expression for its transition kernel in terms of P and π.

2 Action-Value Function

Consider a discounted finite Markov Decision Process $(\mathcal{S}, \mathcal{A}, \mathcal{R}, P, \lambda)$ with discount factor $\lambda \in [0, 1)$. Throughout, $\mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ is equipped with the supremum norm $\|Q\|_\infty := \max_{(s,a)\in\mathcal{S}\times\mathcal{A}} |Q(s,a)|$, under which it is a complete metric space.

In the lectures, the value function v^π of a policy π was introduced. In this question, we develop the analogous state–action value function and its corresponding Bellman operators.

For a policy $\pi \in \Pi^{HR}$, define the state–action value function $Q^\pi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ by

$$Q^\pi(s,a) := \mathbb{E}^\pi_{s,a}\left[\sum_{n=0}^{\infty} \lambda^n r(S_n, A_n)\right],$$

where $\mathbb{E}^\pi_{s,a}$ denotes expectation when the process starts in $S_0 = s$, takes action $A_0 = a$ at time 0, and follows π from time 1 onward.

(a) **Bellman equation for Q^π.**

Let $\pi = \{\phi, \phi, \dots\}$ be a **stationary** randomized policy for some $\phi : \mathcal{S} \to \Delta(\mathcal{A})$. Write down $Q^\pi(s,a)$ as a recursive expression (Bellman equation).

(b) **Q-Operator.**

Let $\pi = \{\phi, \phi, \ldots\}$ be a stationary randomized policy as in (a). Define the operator $L_\pi^Q : \mathbb{R}^{|\mathcal{S}||\mathcal{A}|} \to \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ by

$$(L_\pi^Q Q)(s,a) := Q^\pi(s,a).$$

Write L_π^Q in vector form. Show that it is a contraction mapping with modulus λ with respect to $\|\cdot\|_\infty$.

Using the Banach Fixed Point Theorem, conclude that L_π^Q has a unique fixed point.

(c) **Optimal state–action value function.**

Define the optimal state–action value function by

$$Q_{\infty,\lambda}^*(s,a) := \sup_{\pi \in \Pi^{HR}} Q^\pi(s,a).$$

Consider the operator L^Q defined by

$$(L^Q Q)(s,a) := r(s,a) + \lambda \sum_{s' \in \mathcal{S}} P(s' \mid s,a) \max_{a' \in \mathcal{A}} Q(s',a').$$

(i) Show that $Q_{\infty,\lambda}^*$ satisfies the fixed-point equation

$$Q_{\infty,\lambda}^* = L^Q Q_{\infty,\lambda}^*.$$

(ii) Show that L^Q is a contraction mapping with modulus λ with respect to $\|\cdot\|_\infty$. Hence, using the Banach Fixed Point Theorem, conclude that $Q_{\infty,\lambda}^*$ is the unique fixed point of L^Q.

(d) **Connection between $v_{\infty,\lambda}^*$ and $Q_{\infty,\lambda}^*$.**

Show that

$$v_{\infty,\lambda}^*(s) = \max_{a \in \mathcal{A}} Q_{\infty,\lambda}^*(s,a).$$

Furthermore, show that

$$Q_{\infty,\lambda}^*(s,a) = r(s,a) + \lambda \sum_{s' \in \mathcal{S}} P(s' \mid s,a) v_{\infty,\lambda}^*(s').$$

(e) **Optimal action and optimal policy.**

Using the optimal state–action value function, define

$$\pi^*(s) \in \arg\max_{a \in \mathcal{A}} Q_{\infty,\lambda}^*(s,a).$$

Show that

$$v_{\infty,\lambda}^*(s) = Q_{\infty,\lambda}^*(s, \pi^*(s))$$

and explain why π^* is an optimal policy.